Mechanical waves and distant effects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel-time reasoning while distinguishing waves from the motion of material.

SUCCESS CRITERIA - I can...

1. Identify the medium carrying the wave.
2. Use speed and distance to estimate travel time.
3. Explain why a simple travel time does not predict damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “medium” and “wave” to explain your answer. Sentence starters are optional.

medium: the material through which a mechanical wave travels

wave: a disturbance that can carry energy and information

signal: a disturbance that carries information

01 Warm-up: bring back an earlier idea

A converging lens has f=**15 cm** and d_o=**6 cm**.
1. Use **1**/f=**1**/d_o **+1**/d_i and m=-d_i/d_o.
2. Find d_i, magnification and image kind.

Word help: lens = a transparent element that changes light-ray directions

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

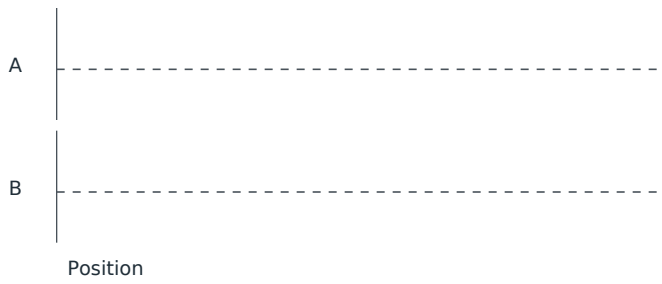
02 Worked example: explain the prediction

Reference: d = v Δt; Δt = d/v.
Use the supplied wave speed for its medium and wave type. These classroom calculations are not real hazard forecasts.

Worked reasoning: In a simplified description, a wave amplitude falls from **4** to **2** units while intensity is proportional to amplitude squared. Intensity becomes one quarter, not one half.

Your turn: Draw the two waves and two corresponding relative-intensity bars.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

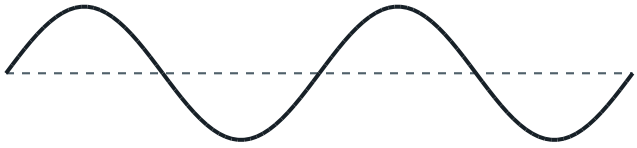


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw a disturbance traveling through** water and **mark local water motion**.
2. Why does wave arrival not mean the original water crossed the whole ocean?

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



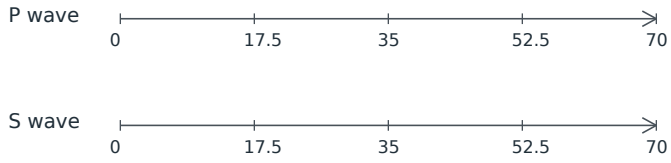
Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A simplified seismic calculation uses P-wave speed **6 km/s** and S-wave speed **3.5 km/s**. A station is **210 km** from the source. **Calculate each arrival time** and the S-P delay. **Mark both arrivals** on the timeline.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is wave arrival time alone not enough to **predict local damage**? **Name other measurements or observations** you would need.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

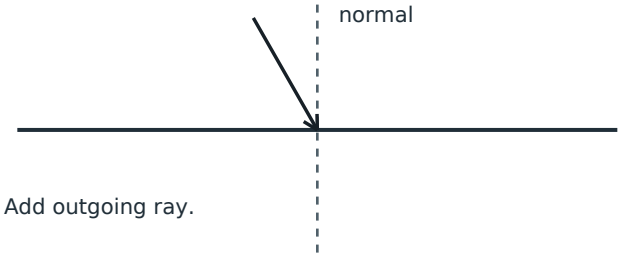
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Light moves from a transparent medium with n=**1.7** into air (n=**1**). **Find critical angle** using sin θ_c=**1**/n. **Describe incidence above it** in this ideal case. **Label the media**, normal and ray directions.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add outgoing ray.

Marzano check - circle: 1 2 3

07 Exit check

A simplified shallow-water calculation uses v=√(gh). With g=**10** and h=**100 m**, find speed and time for **10 km**. **State a limitation**.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



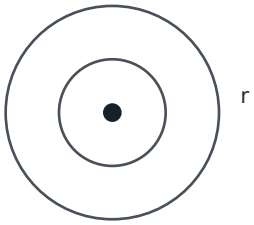
Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A signal has intensity **8 W/m²** at **10 m** from an ideal point source.
1. **Find intensity** at **30 m** and total radiated power.
2. **Explain one limitation** for ground waves.

Word help: ideal = using stated simplifying conditions

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Label distances and compare intensity.

Marzano check - circle: 1 2 3